MTH 201 Chapter 2: Integration Techniques

2.4 Partial Fractions Worksheet

1. Compute $\int \frac{3x}{x^2-2x-8}\,dx$

We know that $\frac{3x}{x^2-2x-8} = \frac{3x}{(x-4)(x+2)} = \frac{A}{(x-4)} + \frac{B}{(x+2)}$. It must be the case that $3x = A(x+2) + B(x-4)$.

If $x=-2$, then $-6=-6B \quad \Rightarrow \quad B=1$

If $x=4$, then $12=6A \quad \Rightarrow \quad A=2$

This means that:

$$\int \frac{3x}{x^2-2x-8}\,dx = \int \left(\frac{2}{x-4} + \frac{1}{x+2}\right)dx = 2\ln|x-4| + \ln|x+2| + C$$

2. Compute $\int_2^3 \frac{x-4}{x^3-x^2}\,dx$

We know that $\frac{x-4}{x^3-x^2} = \frac{x-4}{x^2(x-1)} = \frac{A}{x} + \frac{B}{x^2} + \frac{C}{x-1}$. It must be the case that $x-4 = Ax(x-1) + B(x-1) + Cx^2$.

If $x=0$, then $-4=-B \quad \Rightarrow \quad B=4$

If $x=1$, then $-3=C$

Therefore if $x=4$, then $0=12A+3(4)+16(-3) \quad \Rightarrow \quad 36=12A \quad \Rightarrow \quad A=3$

This means that:

$$\int_2^3 \frac{x-4}{x^3-x^2}\,dx = \int_2^3 \left(\frac{3}{x} + \frac{4}{x^2} - \frac{3}{x-1}\right)dx = \left(3\ln|x| - \frac{4}{x} - 3\ln|x-1|\right)\Big|_2^3$$

$$= \left(3\ln(3) - \frac{4}{3} - 3\ln(2)\right) - \left(3\ln(2) - 2 - 3\ln(1)\right)$$

$$= 3\ln(3) - 6\ln(2) + \frac{2}{3}$$

3. Compute $\int \frac{2x^2+5x+12}{x^3+2x^2+x+2}\,dx$

We know that $\frac{2x^2+5x+12}{x^3+2x^2+x+2} = \frac{2x^2+5x+12}{(x^2+1)(x+2)} = \frac{Ax+B}{x^2+1} + \frac{C}{x+2}$. It must be the case that

$2x^2+5x+12 = (Ax+B)(x+2) + C(x^2+1) = (A+C)x^2 + (2A+B)x + (2B+C)$. Therefore:

$$\begin{cases} A+C=2 \\ 2A+B=5 \\ 2B+C=12 \end{cases} \quad \Rightarrow \quad A=0,\ B=5,\ C=2$$

This means that:

$$\int \frac{2x^2+5x+12}{x^3+2x^2+x+2}\,dx = \int \left(\frac{5}{x^2+1} + \frac{2}{x+2} \right) dx = 5\tan^{-1}(x) + 2\ln|x+2| + C$$

4. Compute $\int_{-5}^{-1} \frac{x+2}{x^3-2x^2+x}\,dx$

We know that $\frac{x+2}{x^3-2x^2+x} = \frac{x+2}{x(x-1)^2} = \frac{A}{x} + \frac{B}{(x-1)} + \frac{C}{(x-1)^2}$. It must be the case that

$x+2 = A(x-1)^2 + Bx(x-1) + Cx$.

If $x=0$, then $2=A$
If $x=1$, then $3=C$
Therefore if $x=2$, then $4=(2)+2B+2(3) \Rightarrow -4=2B \Rightarrow B=-2$

This means that:

$$\int_{-5}^{-1} \frac{x+2}{x^3-2x^2+x}\,dx = \int_{-5}^{-1} \left(\frac{2}{x} - \frac{2}{x-1} + \frac{3}{(x-1)^2} \right) dx = \left(2\ln|x| - 2\ln|x-1| - \frac{3}{x-1} \right)\Bigg|_{-5}^{-1}$$

$$= \left[2\ln(1) - 2\ln(2) + \frac{3}{2} \right] - \left[2\ln(5) - 2\ln(6) + \frac{1}{2} \right]$$

$$= 2\ln(6) - 2\ln(5) - 2\ln(2) + 1$$

$$= 2\ln\left(\frac{3}{5}\right) + 1$$

5. Compute $\int \frac{x^3+4x^2+2x-1}{x^2+x-2}\,dx$

For this indefinite integral we must start by doing polynomial long division. The polynomial long division (which is not shown here) yields: $\frac{x^3+4x^2+2x-1}{x^2+x-2} = x+3+\frac{x+5}{x^2+x-2} = x+3+\frac{x+5}{(x+2)(x-1)}$.

We know that $\frac{x+5}{(x+2)(x-1)} = \frac{A}{(x+2)} + \frac{B}{(x-1)}$. It must be the case that $x+5 = A(x-1)+B(x+2)$.

If $x=-2$, then $3=-3A \quad \Rightarrow \quad A=-1$

If $x=1$, then $6=3B \quad \Rightarrow \quad B=2$

This means that:

$$\int \frac{x^3+4x^2+2x-1}{x^2+x-2}\,dx = \int \left(x+3-\frac{1}{x+2}+\frac{2}{x-1} \right) dx = \frac{1}{2}x^2+3x-\ln|x+2|+2\ln|x-1|+C$$

6. Compute $\int \frac{5}{\left(x^2+4\right)\left(x-1\right)}dx$

We know that $\frac{5}{\left(x^2+4\right)\left(x-1\right)}=\frac{Ax+B}{x^2+4}+\frac{C}{x-1}$. It must be the case that

$5=\left(Ax+B\right)\left(x-1\right)+C\left(x^2+4\right)=\left(A+C\right)x^2+\left(-A+B\right)x+\left(-B+4C\right)$. Therefore:

$$\begin{cases} A+C=0 \\ -A+B=0 \\ -B+4C=5 \end{cases} \Rightarrow \quad A=-1,\ B=-1,\ C=1$$

This means that:

$$\begin{aligned}
\int \frac{5}{\left(x^2+4\right)\left(x-1\right)}dx &= \int\left(\frac{-x-1}{x^2+4}+\frac{1}{x-1}\right)dx = \int\left(-\frac{x}{x^2+4}-\frac{1}{x^2+4}+\frac{1}{x-1}\right)dx \\
&= -\int \frac{x}{x^2+4}dx - \int \frac{1}{x^2+4}dx + \int \frac{1}{x-1}dx \\
&= -\int \frac{x}{x^2+4}dx - \int \frac{1/4}{x^2/4+1}dx + \int \frac{1}{x-1}dx \\
&= -\int \frac{x}{x^2+4}dx - \frac{1}{4}\int \frac{1}{\left(x/2\right)^2+1}dx + \int \frac{1}{x-1}dx
\end{aligned}$$

On the first integral we use the following substitution: Let $u=x^2+4 \Rightarrow du=2xdx \Rightarrow \frac{1}{2}du=xdx$.

On the second integral we use the following substitution: Let $v=\frac{x}{2} \Rightarrow dv=\frac{1}{2}dx \Rightarrow 2dv=dx$

So then we have that:

$$\begin{aligned}
\int \frac{5}{\left(x^2+4\right)\left(x-1\right)}dx &= -\int \frac{x}{x^2+4}dx - \frac{1}{4}\int \frac{1}{\left(x/2\right)^2+1}dx + \int \frac{1}{x-1}dx = -\frac{1}{2}\int \frac{1}{u}du - \frac{1}{4}\cdot 2\int \frac{1}{\left(v\right)^2+1}dv + \int \frac{1}{x-1}dx \\
&= -\frac{1}{2}\ln\left|u\right| - \frac{1}{2}\tan^{-1}\left(v\right) + \ln\left|x-1\right| + C \\
&= -\frac{1}{2}\ln\left|x^2+4\right| - \frac{1}{2}\tan^{-1}\left(\frac{x}{2}\right) + \ln\left|x-1\right| + C
\end{aligned}$$

7. Find $\int \frac{x^4+x^3+2x^2+2x+1}{x^3+x^2}dx$.

For this indefinite integral we must start by doing polynomial long division. The polynomial long division (which is not shown here) yields: $\frac{x^4+x^3+2x^2+2x+1}{x^3+x^2} = x+\frac{2x^2+2x+1}{x^3+x^2} = x+\frac{2x^2+2x+1}{x^2(x+1)}$.

Now, we wish to find A, B, and C so that they satisfy: $\frac{2x^2+2x+1}{x^2(x+1)} = \frac{A}{x+1}+\frac{B}{x}+\frac{C}{x^2}$.

Now, we solve for A, B, and C: $\frac{2x^2+2x+1}{x^2(x+1)} = \frac{A}{x+1}+\frac{B}{x}+\frac{C}{x^2} \Rightarrow 2x^2+2x+1 = Ax^2+Bx(x+1)+C(x+1)$.
If we let $x=0$ we see that: $1=C$. If we let $x=-1$ we see that: $1=A$. We cannot make A and C disappear with a particular value of x. So, we let $x=1$ and we use what we know to obtain:
$5=A+2B+2C \Leftrightarrow 5=1+2B+2(1) \Leftrightarrow 2=2B \Leftrightarrow B=1$.

This means that:

$$\int \frac{x^4+x^3+2x^2+2x+1}{x^3+x^2}dx = \int\left(x+\frac{2x^2+2x+1}{x^2(x+1)}\right)dx$$

$$= \int\left(x+\frac{1}{x+1}+\frac{1}{x}+\frac{1}{x^2}\right)dx = \frac{1}{2}x^2+\ln|x+1|+\ln|x|-\frac{1}{x}+C\ .$$